

IAS PRELIMS 2020

STATE BY STATE



TAMILNADU

DEMOGRAPHICAL

GEOGRAPHICAL

ENVIRONMENTAL

CULTURAL

MODULE - 16

1

HIGHS AND LOWS



1

Tamil was the first language to be recognised as a classical language of India.

2

Marina Beach in Chennai is the longest natural urban beach in the country.

3

Three Biosphere Reserves under UNESCO's WBNR, highest in India.

4

Tamil Nadu Defence Industrial Corridor is the country's second defence industrial corridor after Uttar Pradesh.

5

Yarn Bowl of India (Produces 40% of total Yarn).

6

Home to the highest number of factories in India.

7

Highest number of operational SEZs in India.

8

Automobile Capital of India. And also, among top 10 manufacturing hubs in the world.

STATE BY STATE ... TAMIL NADU

2

BOUNDARIES

ARABIAN SEA

BAY OF BENGAL

LAKSHADWEEP

ANDHRA
PRADESH

3

KARNATAKA

2

Chennai

Puducherry

Karaikal

TAMIL NADU

KERALA

1

SRI LANKA

● STATE BOUNDARY

- Bordered by 3 States and 1 UT (Puducherry).
- Shares a maritime border with Sri Lanka.

KARNATAKA

PRADESH

ARABIAN SEA

Nilgiri Mountains

TAMIL NA

KERALA

Kanyakumari

INDIAN OCEAN

Gulf of Mannar

Palk Bay

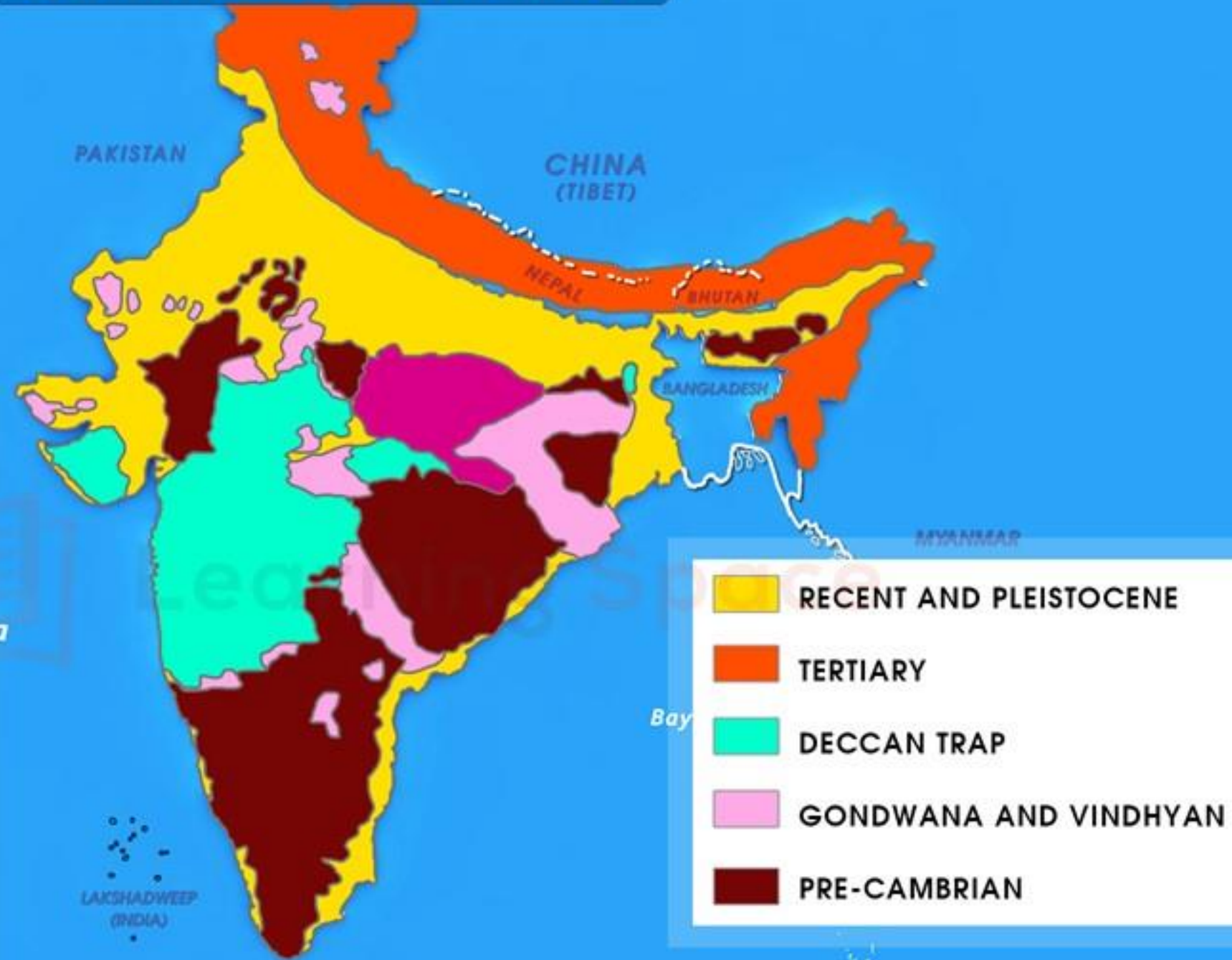
Palk Strait

SRI LANKA

- Bounded by –
 - ✓ Eastern Ghats on the north,
 - ✓ Nilgiri Mountains and Kerala on the West;
 - ✓ Bay of Bengal in the East;
 - ✓ Gulf of Mannar and the Palk Strait on the Southeast;
 - ✓ Indian Ocean on the South.
- Kanyakumari is the southernmost tip of the Indian Peninsula. Meeting point of Arabian Sea, Bay of Bengal, and Indian Ocean.

LAKSHADWEEP

- Pre-Cambrian rocks with mineral assemblages of high temperature and pressure are well exposed in Tamil Nadu.
 - ✓ Oldest rocks (formed in the Pre-Cambrian era – about 4 billion years ago).
 - ✓ Formed due to solidification of molten magma.
 - ✓ Known as the 'Basement Complex' as they are the oldest and forms the base for new layers.



STATE BY STATE ... TAMIL NADU

3

RELIEF FEATURES

ARABIAN SEA

KARNATAKA

ANDHRA
PRADESH

BAY OF BENGAL

KERALA

- Western, Southern and the North-western parts are hilly and rich in vegetation.
- Eastern parts are fertile coastal plains and the northern parts are a mix of hills and plains.
- Central and the south central regions are arid plains and receive less rainfall than the other regions.

STATE BY STATE ... TAMIL NADU

3

RELIEF FEATURES

ARABIAN SEA

KARNATAKA

ANDHRA
PRADESH

BAY OF BENGAL

Nilgiri Hills

Kunda Hills

Velliangiri
Mountains

Lower Cauvery Plain

Palani Hills

Anaimalai

Kambam Valley

Upper Vaigai
Valley

Varushanad
Hills

Agasthyamalai

Javadi
Hills

Shevaroy
Hills

Gingee
Hills

Chitteri
Hills

Kalrayan
Hills

Pachaimalai Hills

- Anai Peak in the Anaimalai Hills, is the highest mountain in Peninsular India.
- Lower peaks of the Eastern Ghats and their outliers — locally called the Javadi, Kalrayan, and Shevaroy hills — run through the centre of the region.

INDIA

MAHARASHTRA

KARNATAKA

TAMIL NADU

Javadi Hills

Tamil Nadu

Shevaroy Hills

Tamil Nadu

Pachamalai Hills

Tamil Nadu

Sirumali Hills

Tamil Nadu

Palani Hills

Tamil Nadu

Cardamom Hills

Kerala & Tamil Nadu

Nilgiri Mountains

Tamil Nadu, Kerala & Karnataka

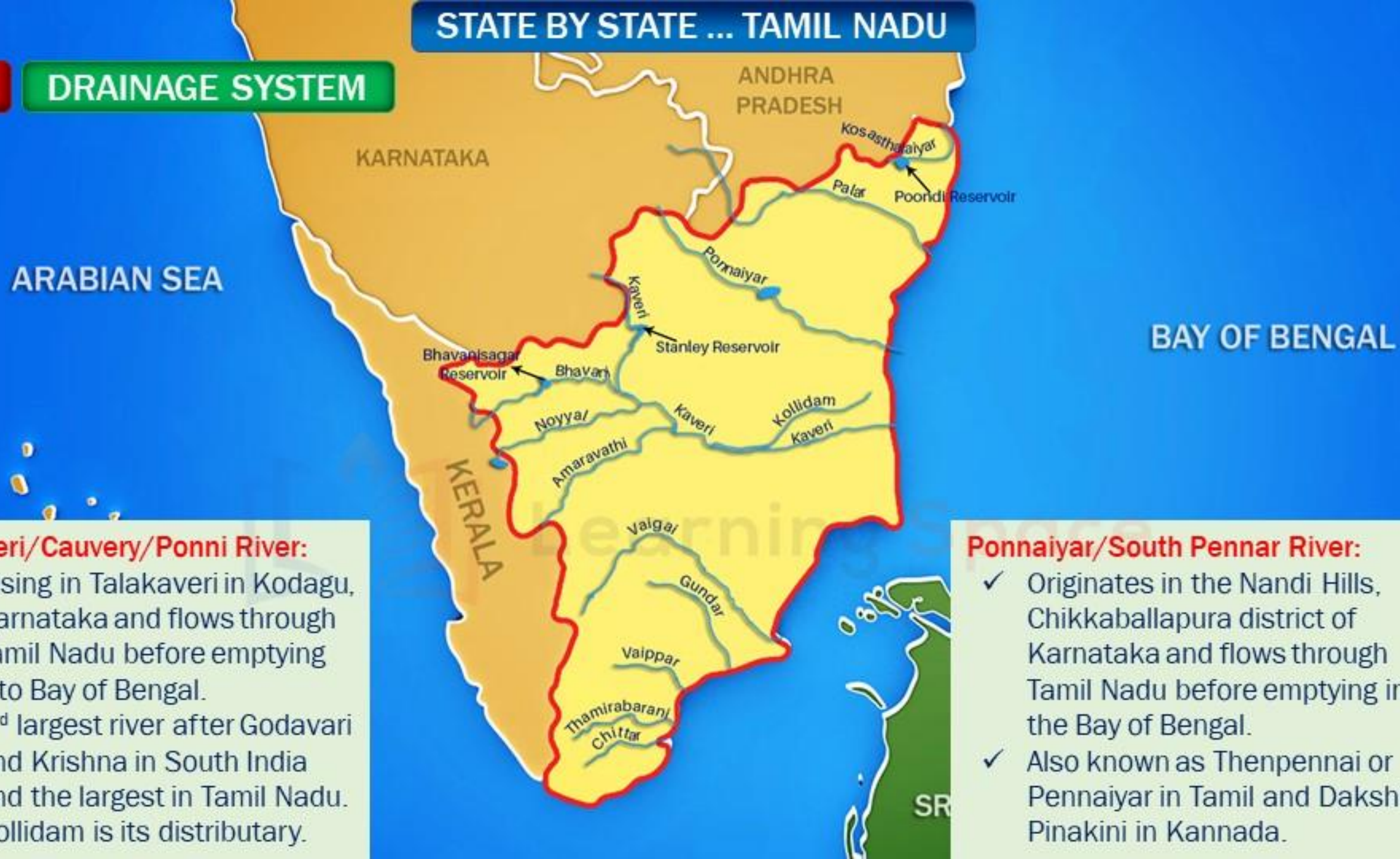
Anaimalai Hills

Tamil Nadu & Kerala

Agasthyamalai Hills

Tamil Nadu & Kerala

- Highest peaks of Western Ghats run along the western border.
- Western Ghats and the Eastern Ghats meet at the Nilgiri hills.
- Western Ghats traverse the entire western border with Kerala, effectively blocking much of the rain bearing clouds of the south west monsoon from entering the state.



Kaveri/Cauvery/Ponni River:

- Rising in Talakaveri in Kodagu, Karnataka and flows through Tamil Nadu before emptying into Bay of Bengal.
- 3rd largest river after Godavari and Krishna in South India and the largest in Tamil Nadu.
- Kollidam is its distributary.

Ponnaiyar/South Pennar River:

- ✓ Originates in the Nandi Hills, Chikkaballapura district of Karnataka and flows through Tamil Nadu before emptying into the Bay of Bengal.
- ✓ Also known as Thenpennai or Pennaiyar in Tamil and Dakshina Pinakini in Kannada.

**Palar River:**

- ✓ Rises in Nandi Hills in Karnataka.
- ✓ Water from Palar anicut is diverted to Poondi reservoir in the Kosasthalaiyar River basin and to Chembambakkam Lake located in the Adayar River basin.
- ✓ These two reservoirs are major water supply points to Chennai.
- ✓ After commissioning of the Telugu Ganga project to supply Krishna River water to the Chennai city, the dependence on Palar river water has reduced drastically.

ARABIAN SEA

BAY OF BENGAL



- **Amaravati River:** Begins at the Tamil Nadu-Kerala border between the Anamalai Hills and the Palani Hills. Longest tributary of Kaveri River.
- **Noyyal River:** Rises from the Vellingiri hills in the Western Ghats in Tamil Nadu. Tributary of Kaveri River.
- **Bhavani River:** Originates from Nilgiri hills, enters the Silent Valley National Park in Kerala and flows back towards Tamil Nadu to join Kaveri River.

- **Vaigai River:** Originates in Varushanad Hills, the Periyar Plateau of the Western Ghats range, and flows northeast through the Kambam Valley, which lies between the Palani Hills to the north and the Varushanad Hills to the south. Empties into the Palk Bay.
- **Thamirabarani River:** Originates from the Agastyarkoodam peak of the Western Ghats.

ARABIAN SEA

ANDHRA
PRADESH

Krishnagiri

Mettur

Bhavani Sagar

Amaravathi

Vaigai

Sathanur

BAY OF BENGAL

All these dams are part of irrigation projects of Tamil Nadu.

Dam

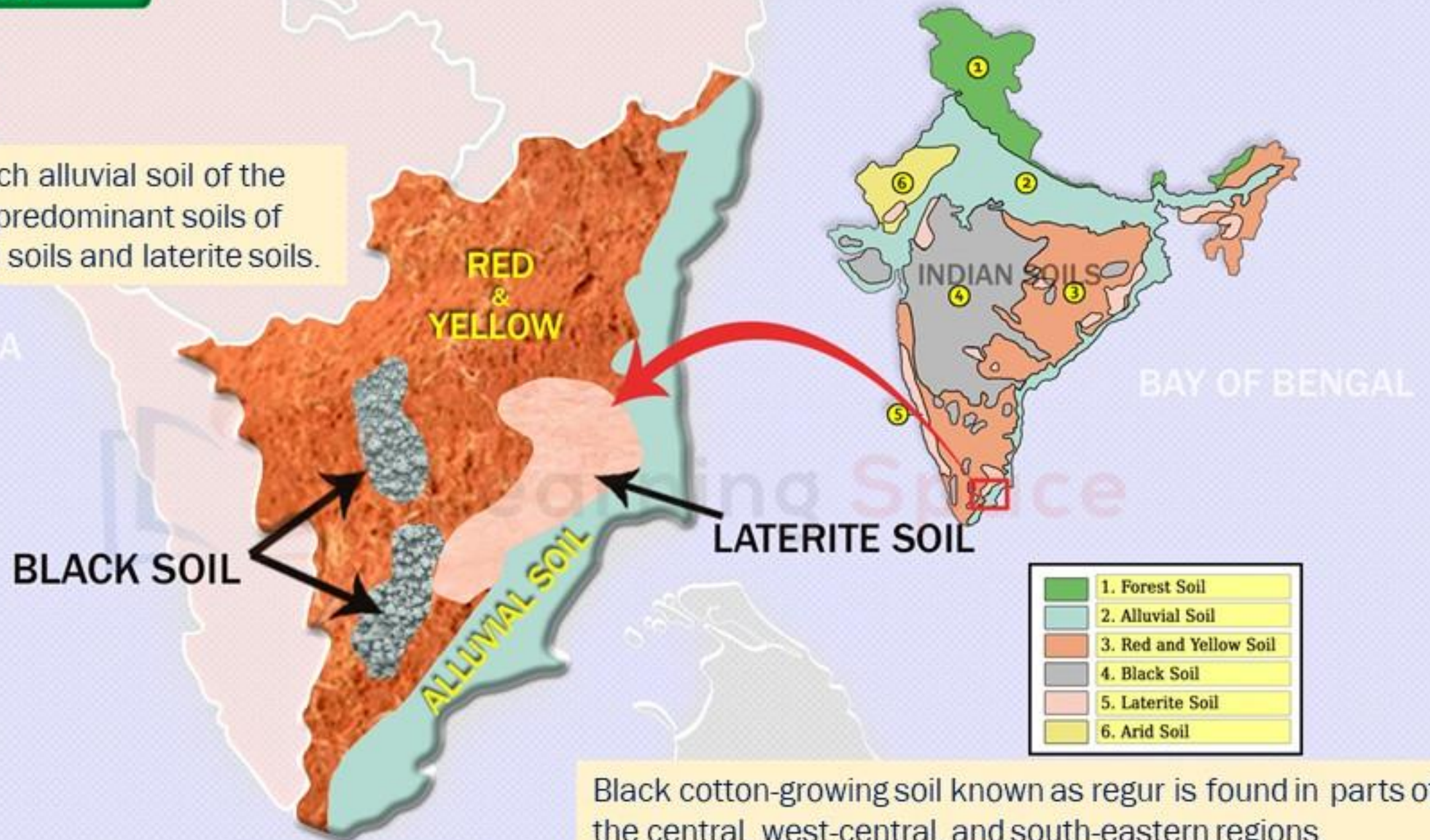
- Amaravathi
- Mettur
- Sathanur
- Vaigai
- Krishnagiri
- Bhavani Sagar

River

- Amaravathi River (Cauvery Basin)
- Cauvery River
- Ponniyar River
- Vaigai River
- Ponniyar River
- Bhavani River

Mettur dam: Nearly goes dry during certain parts of the year, when water is most needed by the farmers and general public of Karnataka and Tamil Nadu.

Apart from the rich alluvial soil of the river deltas, the predominant soils of the state are red soils and laterite soils.



Black cotton-growing soil known as regur is found in parts of the central, west-central, and south-eastern regions.

STATE BY STATE ... TAMIL NADU

6

RAINFALL

- Mostly dependent on monsoon rains and thereby prone to droughts.
- South west monsoon (32%) from June to September, with strong southwest winds.
- North east monsoon (48%) from October to December, with dominant northeast winds.

Climate: Ranges from dry sub-humid to semi-arid.



ARABIAN SEA

ANDHRA
PRADESH

KARNATAKA

KERALA

SRI LANKA

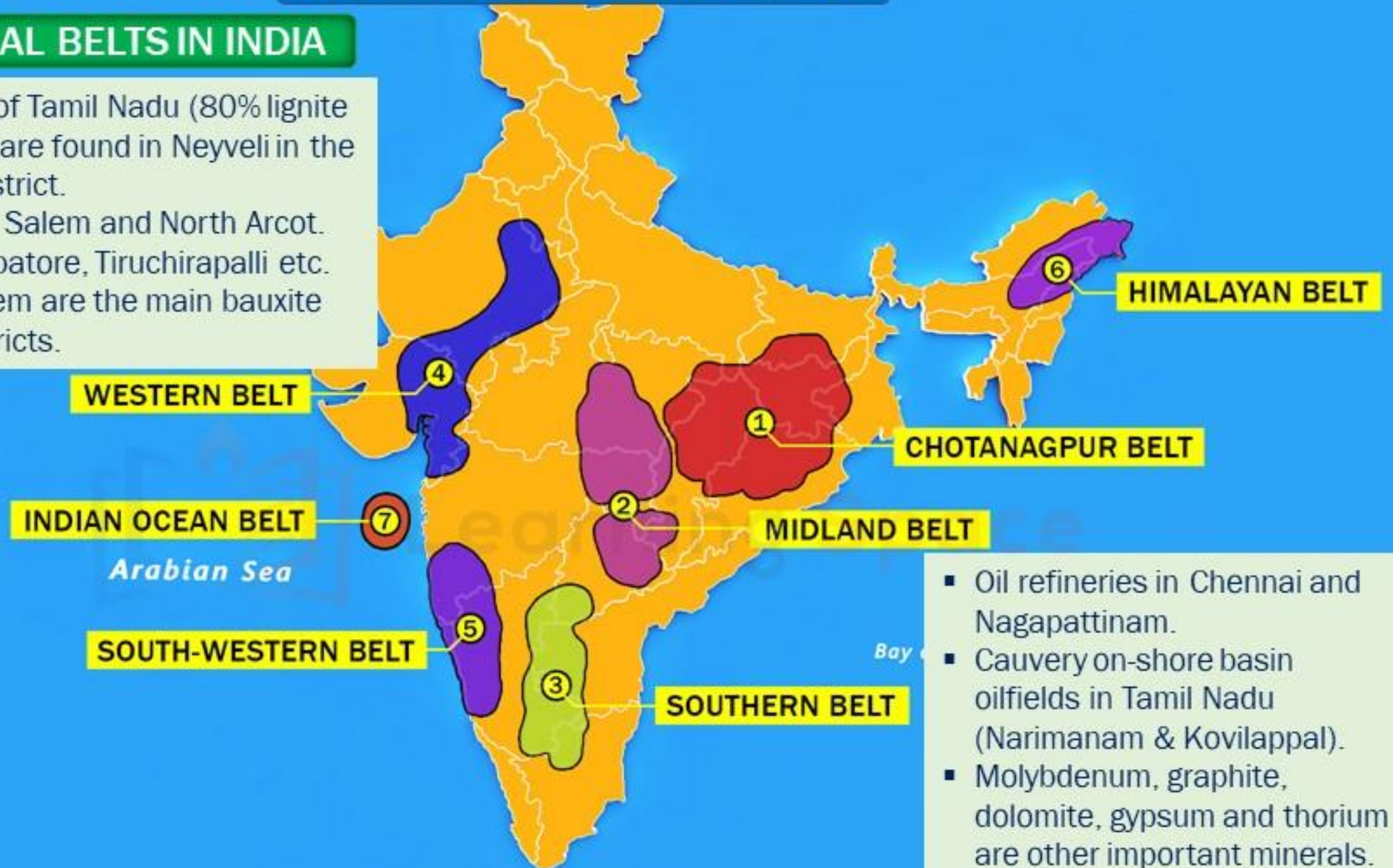
8 Cauvery delta districts including Thanjavur (the rice bowl of Tamil Nadu) categorised as a special protected agricultural zone and permission will not be given for hydrocarbon projects.

- Tamil Nadu is divided into seven agro-climatic zones.
- Principal food crops are rice, jowar, bajra, ragi, and pulses.
- Cash crops include cotton, sugarcane, oilseeds, coffee, tea, rubber, coconut and chillies.
- Important horticultural products are bananas and mangoes.
- Farmers shifted from maize to cotton due to Fall Armyworm attack.

	Rice		Ragi
	Sugarcane		Oil Seeds
	Groundnut		Potato
	Betel Leaf		Mango
	Pulses		Chilli
	Cotton		Other Crops

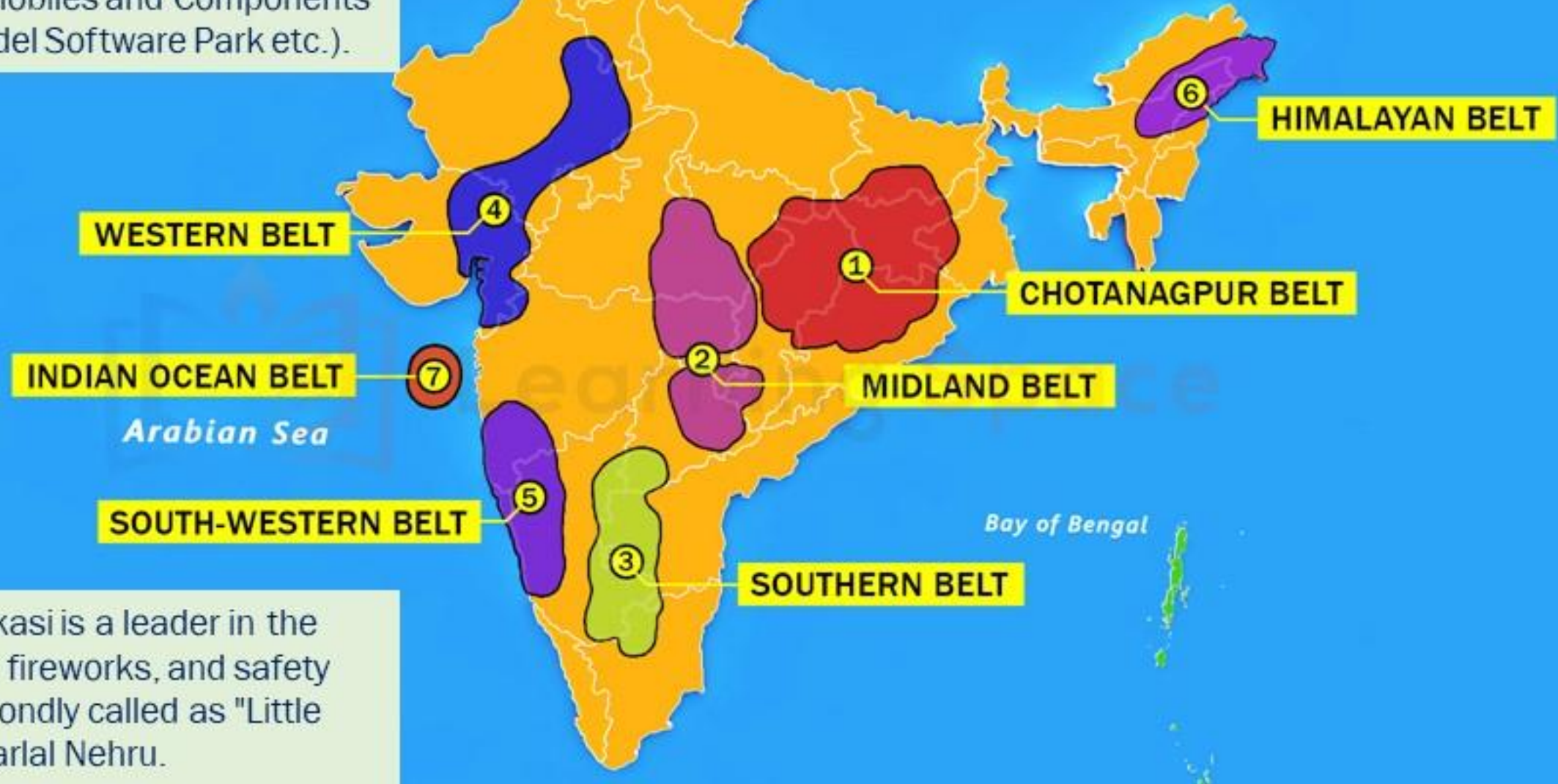
MINERAL BELTS IN INDIA

- Coal deposits of Tamil Nadu (80% lignite coal reserves) are found in Neyveli in the South Arcot district.
- Magnetite ore: Salem and North Arcot.
- Iron-ore: Coimbatore, Tiruchirapalli etc.
- Nilgiri and Salem are the main bauxite producing districts.



- Oil refineries in Chennai and Nagapattinam.
- Cauvery on-shore basin oilfields in Tamil Nadu (Narimanam & Kovilappal).
- Molybdenum, graphite, dolomite, gypsum and thorium are other important minerals.

Industries: Known as a major exporter of Leather and Leather Goods, Textiles and Garments, Automobiles and Components and Software (Tidel Software Park etc.).



The town of Sivakasi is a leader in the areas of printing, fireworks, and safety matches. It was fondly called as "Little Japan" by Jawaharlal Nehru.

STATE BY STATE ... TAMIL NADU

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INFRASTRUCTURE ... POWER PLANTS

ARABIAN SEA

ANDHRA
PRADESH

North Chennai

Ennore

Mettur

Kalpakkam

BAY OF BENGAL

Neyveli

KERALA

Kudankulam

Muppandal

Tuticorin

- NUCLEAR POWER PLANT
- THERMAL POWER STATION
- WIND FARM

- **Kudankulam:** Largest nuclear power station in India.
- **Kalpakkam Nuclear Power Plant/Madras Atomic Power Station (MAPS):** India's first fully indigenously constructed nuclear power station. Located at Kalpakkam.

ARABIAN SEA

ANDHRA
PRADESH

Ennore Port

Chennai Port

- **Kamarajar Port Limited (formerly Ennore Port):** Located on the Coromandel Coast. Handles all the coal and ore traffic in Tamil Nadu.
- **Chennai Port (formerly known as Madras Port):** Second largest container port of India, behind Mumbai's Nhava Sheva. An artificial and all-weather port with wet docks.

KERALA

Tuticorin Port

SRI LANKA

- **V. O. Chidambaranar Port Trust (formerly Tuticorin Port Trust):** Second-largest port in Tamil Nadu and fourth-largest container terminal in India.
- **Colachel International Seaport:** Also known as Enayam port, is proposed and located in Colachel, Kanyakumari.

ARABIAN SEA

BAY OF BENGAL

Bengaluru

Tiruvallur

Ennore

Chennai

Tiruchi

Nagapattinam

Madurai

Ramanathapuram

Tuticorin

SRI LANKA

- India's first east coast LNG import terminal at Ennore.
- Part of India's plan to raise the share of natural gas in the country's energy basket to 15% by 2030 from current 6.2%.
- Pipeline from the terminal will go up to Madurai, Trichy and Tuticorin in Tamil Nadu and branch out to Bengaluru via Hosur in Karnataka.

STATE BY STATE ... TAMIL NADU

9

INFRASTRUCTURE ...
GOLDEN QUADRILATERAL



9

INFRASTRUCTURE ... ECONOMIC CORRIDORS

STATE BY STATE ... TAMIL NADU





STATE BY STATE ... TAMIL NADU

10

NATIONAL PARKS

ARABIAN SEA

ANDHRA
PRADESH

KARNATAKA

Guindy NP

Mudumalai NP

Mukurthi NP

Anamalai NP

Gulf of Mannar Marine NP: Protected area consisting of 21 small islands (islets) and adjacent coral reefs in the Gulf of Mannar in the Indian Ocean. Core area of the Gulf of Mannar Biosphere Reserve.

Gulf of Mannar Marine NP

Mukurthi NP: Created to protect its keystone species, the Nilgiri tahr. Part of Nilgiri Biosphere Reserve. Also, UNESCO World Heritage Site as part of the Western Ghats.

Mudumalai NP: Lies on the north-western side of the Nilgiri Hills. Also, shares its boundaries with Karnataka and Kerala. Construction activities of the proposed India-based Neutrino Observatory (INO) likely to have significant impacts on the local wildlife.

Anamalai NP/Indira Gandhi NP: Notified as Anamalai Tiger Reserve.

STATE BY STATE ... TAMIL NADU

11

WILDLIFE SANCTUARIES

ARABIAN SEA

ANDHRA
PRADESH

Pulicat

Vedanthangal

Sathyamangalam

Mudumalai

Indira Gandhi (Annamalai)

Point Calimere

Kodiakkarai

Grizzled Squirrel

Mundanthurai

Kanyakumari

Gangaikondan Spotted Deer Sanctuary

Vallanadu Black Buck Sanctuary

WLS - WILDLIFE SANCTUARY

Grizzled Squirrel WLS (GSWS): Also known as Srivilliputhur Wildlife Sanctuary. Established to protect the vulnerable grizzled giant squirrel.

Kanyakumari WLS: Area is a tiger habitat. Seven rivers originate in the forest including the Thamirabarani River and Pahrli River.

Kodiakkarai WLS: A unique exercise of re-habiting deer population started.

Vallanadu WLS: Created for the protection of blackbuck antelope.

STATE BY STATE ... TAMIL NADU

11

WILDLIFE SANCTUARIES

ARABIAN SEA

ANDHRA
PRADESH

Pulicat

Vedanthangal

Sathyamangalam

Mudumalai

Indira Gandhi (Annamalai)

Point Calimere Wildlife and Bird Sanctuary (PCWBS): Protected area along the Palk Strait where it meets the Bay of Bengal at Point Calimere. Famous for large congregations of water birds, especially greater flamingos.

Point Calimere

Kodiakkarai

Vedanthangal Bird Sanctuary: Recently, State forest department has proposed to reduce the buffer zone to allow construction activity.

Grizzled Squirrel

Mundanthurai

Kanyakumari

Gangaikondan Spotted Deer Sanctuary

Vallanadu Black Buck Sanctuary

WLS - WILDLIFE SANCTUARY

SRI LANKA

KERALA

STATE BY STATE ... TAMIL NADU

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RAMSAR WETLANDS

ARABIAN SEA

ANDHRA
PRADESH

BAY OF BENGAL

Point Calimere

Point Calimere Wildlife & Bird Sanctuary: Only site to be declared as a Ramsar Site in Tamil Nadu.

RW - RAMSAR WETLANDS

KERALA

SRI LANKA



STATE BY STATE ... TAMIL NADU

13

TIGER RESERVES

ARABIAN SEA

Sathyamangalam

Mudumalai

Anamalai

Kalakkad-Mundanthurai

TR - TIGER RESERVE

ANDHRA
PRADESH

BAY OF BENGAL

Anamalai: Earlier known as Indira Gandhi Wildlife Sanctuary and National Park (IGWLS&NP) and previously as Anaimalai Wildlife Sanctuary, is a protected area located in the Anaimalai Hills.

Sathyamangalam: Part of Project Tiger and Project Elephant conservation programmes. Man-animal conflicts are common especially with elephants and leopards. Home to indigenous tribal people belonging largely to the Irula tribe (also known as the Urali) and Soliga communities.

Kalakkad Mundanthurai Tiger Reserve (KMTR): Part of Agasthyamalai Biosphere Reserve

ELEPHANT RESERVES (ER)

BRAHMAGIRI-NILGIRI-EASTERN
GHAT LANDSCAPE

MYSORE ER
KARNATAKA

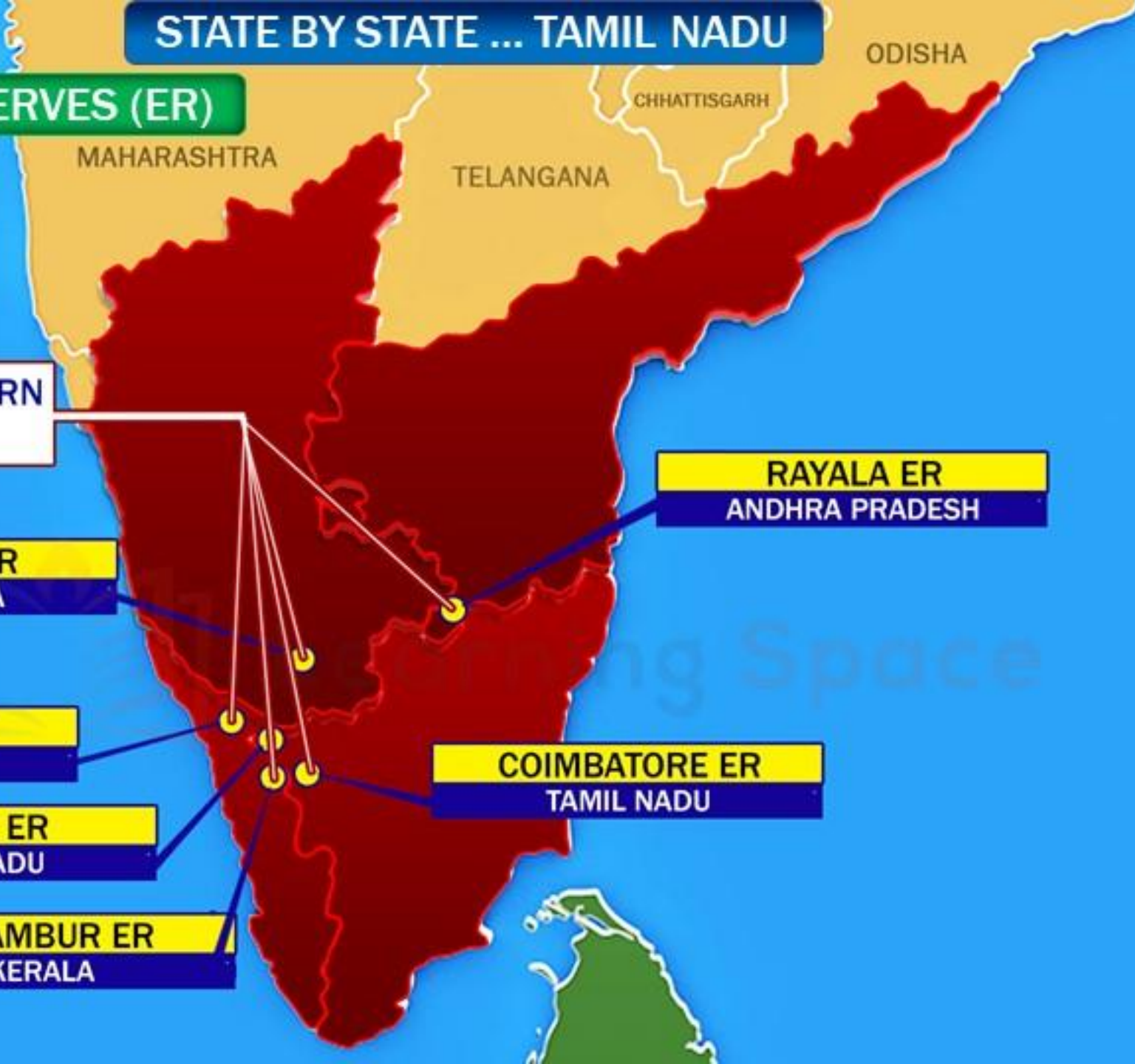
WAYANAD ER
KERALA

NILGIRI ER
TAMIL NADU

NILAMBUR ER
KERALA

COIMBATORE ER
TAMIL NADU

RAYALA ER
ANDHRA PRADESH



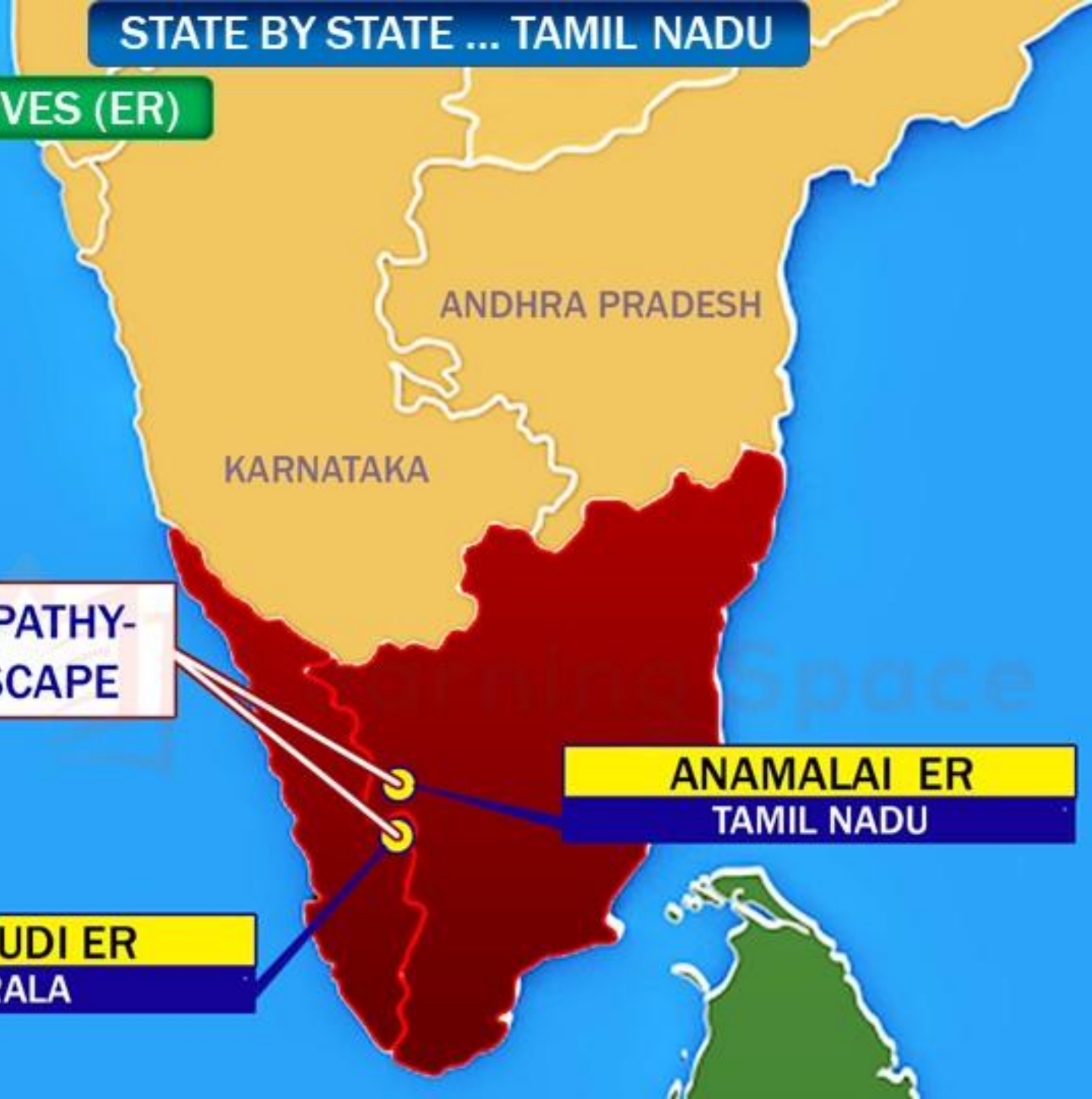
ANAMALAI-NELLIAMPATHY-
HIGH RANGE LANDSCAPE

ANDHRA PRADESH

KARNATAKA

ANAMALAI ER
TAMIL NADU

ANAMUDI ER
KERALA



STATE BY STATE ... TAMIL NADU

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ELEPHANT RESERVES (ER)

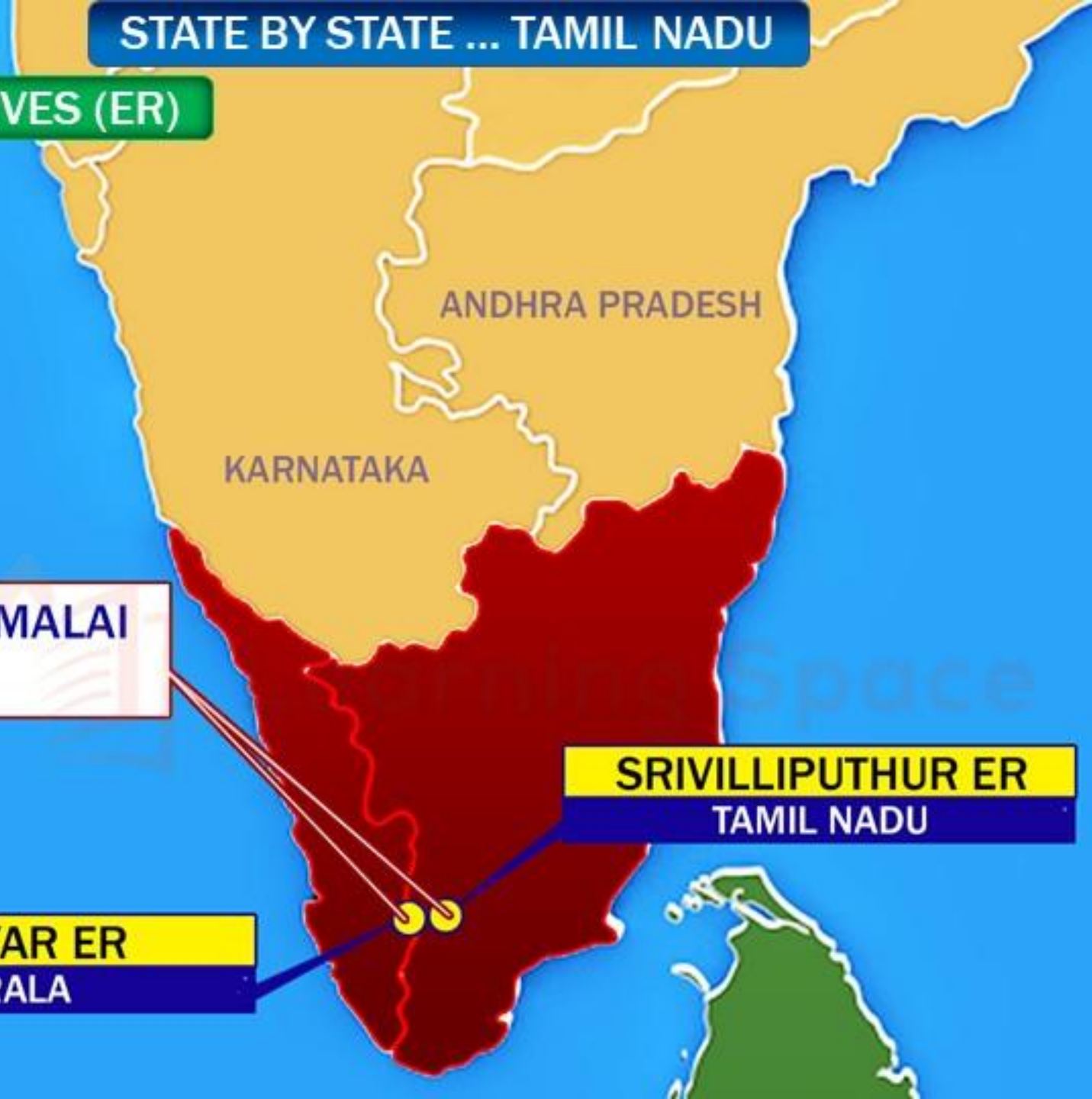
PERIYAR-AGASTHYAMALAI
LANDSCAPE

ANDHRA PRADESH

KARNATAKA

SRIVILLIPUTHUR ER
TAMIL NADU

PERIYAR ER
KERALA



STATE BY STATE ... TAMIL NADU

15

BIOSPHERE RESERVES

ARABIAN SEA

ANDHRA
PRADESH

Nilgiri

- **Agasthyamalai BR:** Under UNESCO's world list of Biosphere Reserves. Became part of World Network of Biosphere Reserves in 2016.
- **Gulf of Mannar BR:** Marine ecosystem with seaweed seagrass communities, coral reefs, salt marshes and mangrove forests. First marine Biosphere Reserve in India.

- **Nilgiri BR:** India's first International Biosphere Reserve. Nilgiri Sub-Cluster is a part of the Western Ghats, which was declared a World Heritage Site by UNESCO in 2012.

KERALA

Gulf of Mannar

Agasthyamalai

SRI LANKA

NOTE: All the three Biosphere Reserves are under UNESCO's World Network of Biosphere Reserves.

STATE BY STATE ... TAMIL NADU

16

UNESCO - WHS

ARABIAN SEA



Nilgiri Mountain Railway



Mandapa



Rathas



Descent of the Ganges

Group of Monuments at Mahabalipuram :

- ✓ Built by Pallava kings
- ✓ Town has gained prominence under the rule of Mamalla.
- ✓ Carved out of rock along the Coromandel coast.
- ✓ The temple town has approximately forty monuments, including the largest open-air bas-relief in the world.
- ✓ Inscribed as a cultural heritage site.

The Mountain Railways of India:

- ✓ represent a collective listing of the Darjeeling Himalayan Railway,
- ✓ the Nilgiri Mountain Railway and
- ✓ the Kalka-Shimla Railway under the UNESCO World Heritage Site.

STATE BY STATE ... TAMIL NADU

16

UNESCO - WHS

ARABIAN SEA

KARNATAKA

ANDHRA
PRADESH

BAY OF BENGAL



Brihadisvara



Gangaikonda Cholapuram



Airavatesvara

SRI LANKA

The Great Living Chola Temples:

- Built during the Chola rule.
- These temples are:
 - ✓ Brihadisvara Temple at Thanjavur,
 - ✓ Temple of Gangaikonda-cholisvaram and
 - ✓ Airavatesvara Temple at Darasuram.

BHARATANATYAM



- *Abhinaya Darpana* by Nandikesvara is one of the main sources of textual material.
- Known to be *ekaharya*, where one dancer takes on many roles in a single performance.
- *Varnam*, the most important composition of the Bharatnatyam repertoire, encompasses both *nritta* and *nritya*.

TANJORE PAINTING



- Thanjavur has a unique place in the history of Indian painting, in that it houses the 11th century Chola wall paintings in the Brihadeeswarar temple and also paintings from the Nayak period.
- Subjects of most paintings are Hindu gods, goddesses, and saints.

BOMMALATTAM



- Puppets from Tamil Nadu known as Bommalattam.
- Combine the techniques of both rod and string puppets.
- Puppets are the largest, heaviest and the most articulate of all traditional Indian marionettes.
- Comes under string puppetry

THERUKOOTHU



- Literally means a “Street play”.
- Performed mostly at the time of annual temple festivals of Mariamman (Rain Goddess) to achieve rich harvest.
- There is a cycle of eight plays based on the life of Draupadi.
- Performance is held in an open air theatre called ‘shabai’.

JALLIKATTU



- A bull taming contest
- Marks the main event of Pongal day.

MAHAMAGAM



- A bathing festival at Kumbakonam in Tamil Nadu
- Celebrated once in 12 years.
- Also called Kumbamela of South.

PVTGs in Tamilnadu: Kattu Nayakans, Kotas, Kurumbas, Irulas, Paniyans, Todas.

KOTAS



- Indigenous to the Nilgiri mountains.
- Study of Todas and Kotas has also been influential in the development of the field of anthropology.
- Women had a greater say in choosing their marriage partners than in any mainstream Indian villages.

IRULAS



- Reside in northern districts of Tamil Nadu and in some parts of Kerala.
- Have their origin from ethnic groups of Southeast Asia and Australia.
- Irula Snake-cooperative has revolutionised the treatment of snake-bites in India and produces enough anti-venom to supply hospitals across the country.

TODAS



- Live in the Nilgiri Mountains
- study of their culture by anthropologists and linguists proved significant in developing the fields of social anthropology and ethnomusicology.
- Toda embroidery products got GI tag.

GRIZZLED GIANT SQUIRREL



- Found in patches of riparian forest along Kaveri River and in hill forests of Karnataka, Tamil Nadu and Kerala. Also found in Sri Lanka.
- Its nests were sighted by researchers for the first time at Pakkamalai Reserve Forests near Gingee (in Eastern Ghats).
- Generally known to nest in foothills of Western Ghats ranging from Chinnar Wildlife sanctuary in Kerala to Anamalai Tiger Reserve and Palani hills in Tamil Nadu and in couple of areas in Eastern Ghats.
- Grizzled Squirrel Wildlife Sanctuary was established to protect vulnerable grizzled giant squirrel.
- Threatened by habitat loss, poaching and also high demand in pet trade.
- IUCN Status: Near Threatened
- Listed under Schedule II of CITES.
- Listed under Schedule I of the Wildlife Protection Act (WPA), 1972.

HUMP-BACKED MAHSEER



- Species of freshwater ray-finned fish and is referred to as tiger of the water.
- Found only in the Cauvery river basin including Pambar, Kabini and Bhavani rivers.
- IUCN Status: Critically Endangered

PEACOCK PARACHUTE SPIDER



- Known habitat is in degraded forests near Nandyal in Andhra Pradesh.
- Recently, it was spotted in a cave in Pakkamalai Reserve Forests in Eastern Ghats near Gingee in Villupuram district of Tamil Nadu.
- Also known as Gooty Tarantula.
- Endemic to India.
- Considered as biological pest controller.
- IUCN Status: Critically Endangered

NEELAKURINJI



- Native to Shola forests in Western Ghats.
- Also seen in Shevroys Hills in Eastern Ghats, Anamalai hills and Agali hills in Kerala and Sanduru hills in Karnataka.
- Blooms once in 12 years.
- No smell or any medicinal value.
- Categorized as endangered species.
- Paliyan tribal people living in Tamil Nadu use this flower bloom as reference to calculate their age.
- Tamil Nadu government has announced Scheme for its protection.

TAMIL YEOMAN



- Locally known as Tamil Maravan, which means 'Tamilian Warrior'.
- Officially declared as the state butterfly to symbolise its rich natural and cultural heritage.
- Aimed at boosting conservation efforts of attractive insects.
- Maharashtra was the first state to officially declare Blue Mormon as its state butterfly.
- Endemic to Western Ghats.

PALLIKARANAI MARSHLANDS



- One of the last few surviving wetlands of South India and the only marshland ecosystem in Chennai city.
- State forest department has announced plans to open this up for the public.
- It is in line with the plans to revive and rejuvenate the wetland.

KEELADI



- Report titled 'Keeladi-An Urban Settlement of Sangam Age on Banks of River Vaigai' published by Tamil Nadu Archaeology Department.
- Recent scientific dates obtained for Keeladi findings push back the date of Tamil-Brahmi script to another century, i.e. 6th century BCE.

DINDIGUL LOCKS



- Known throughout world for their superior quality and durability.
- Abundance of iron in this region is main reason for growth of this industry.
- Government institutions such as prisons, godowns, hospitals, and even temples use these locks instead of other machine-made ones.

KANDANGI SAREES



- Manufactured in entire Karaikudi taluk in Sivaganga district of Tamil Nadu.
- Made up of high quality cotton from Coimbatore.
- Usually worn in summer.
- Manually made using winding machine, shuttle, loom and bobbin.
- It is team effort of families who live in the town of Karaikudi and it forms part of their basic livelihood.

PANCHAMIRTHAM



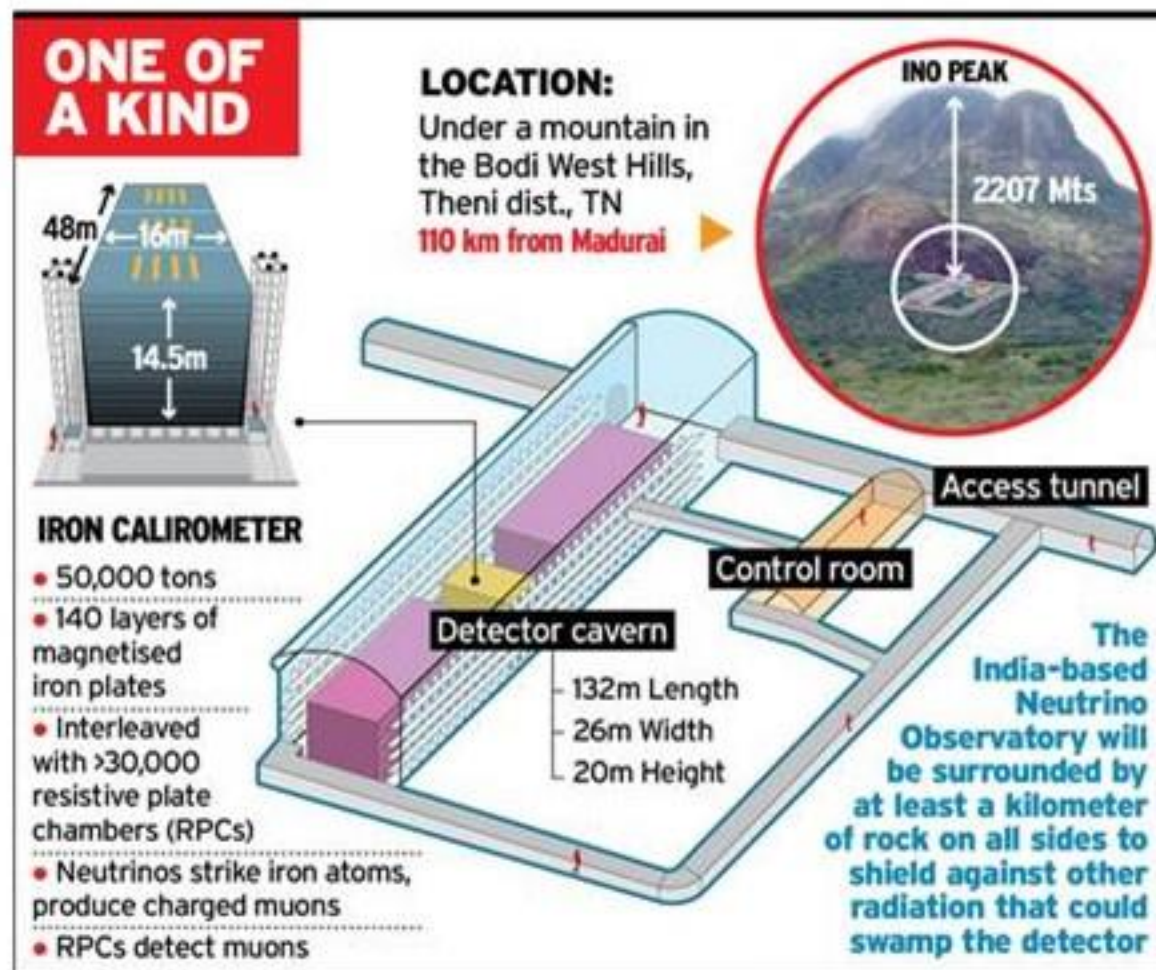
- It is an 'Abhishega prasadam'
- In Tamil, term 'pancha' means five and 'amirtham' means delicacy.
- One of main offerings for Lord Dhandayuthapani Swamy, the presiding deity of Arulmigu Dhandayuthapani Swamy Temple, situated on Palani Hills.

ERODE TURMERIC



- Grown in hot moist conditions prevalent in the area with temperature ranging from 20 degrees to 37.9 degrees Celsius with an average of 600 to 800 mm rainfall in a year.
- Erode is the city in the state of Tamil Nadu.
- Received GI tag after a long and cumbersome process which lasted for about 8 years.

INDIA BASED NEUTRINO OBSERVATORY PROJECT



- Proposed research facility to be constructed close to the Western Ghats.
- Led to concerns over its impact on the eco-sensitive zone.
- High level Working Group, headed by Dr. Kasturirangan, had concluded that the site wasn't in an eco-sensitive region and the NGT has also awarded clearance for its establishment.
- Proposed INO site will study the properties of the elementary particles called neutrino using a larger detector to be constructed in an underground research facility.
- INO will study atmospheric neutrinos only.

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